SyriaTel Customer Churn Prediction

Project Overview

Customer churn is a critical problem for telecom companies, as losing customers directly impacts revenue. This project aims to build a machine learning model to predict customer churn and identify key drivers behind customer attrition.

Business Understanding

The objective of this project is to help SyriaTel reduce customer churn by:

- Identifying customers likely to leave
- Understanding the factors driving churn
- Providing actionable recommendations for retention

Data Understanding

The dataset contains 3,333 customer records with 21 features, including:

- Customer demographics (state, account length)
- Service usage (day, evening, night minutes)
- Account features (international plan, voicemail plan)
- Customer interactions (customer service calls)

Target Variable:

- `churn` (True = customer left, False = customer stayed)

Data Preparation

Key preprocessing steps included:

- Dropping the `phone number` column (identifier, not predictive)
- Encoding categorical variables using one-hot encoding
- Splitting data into training (80%) and testing (20%) sets
- Handling class imbalance using `class_weight='balanced'`

Exploratory Data Analysis (EDA)

Key insights:

- Customers with **more customer service calls** are more likely to churn
- Customers with **international plan** have higher churn rates

---

Results

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.74	0.32	0.70	0.44	0.80
Random Forest	0.92	0.89	0.53	0.66	0.89
Gradient Boosting	0.96	0.95	0.77	0.85	0.89

Final Model:

Gradient Boosting was selected due to its strong balance between precision and recall.

Business Recommendations

- Focus on customers with **high customer service calls**
- Improve service experience to reduce dissatisfaction
- Target **international plan users** with retention offers
- Monitor high-usage customers for potential churn risk

Next Steps

- Deploy the model as an API for real-time predictions
- Build a dashboard for business stakeholders
- Perform hyperparameter tuning for further optimization
- Incorporate additional customer behavior data

Tools & Technologies

- Python (Pandas, NumPy)
- Scikit-learn
- Matplotlib & Seaborn
- Jupyter Notebook

Author

Brian Chairo
Aspiring Data Scientist